SAMUEL PATRONE

Theoretical Particle Physicist and Cosmologist

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WHO I AM

I work at the crucial interface between abstract mathematical models and experimental observations in cosmology and particle physics. My research transforms theoretical predictions into testable signatures that can be probed in laboratories and telescopes. From strengthening axion-like-particle bounds to extracting fundamental physics from gravitational waves, I specialize in making the *invisible universe* experimentally accessible.

EDUCATION

Ph.D. in Physics *2020–2026*
California Institute of Technology | GPA: 4.3/4.0

Koh Fellow (2020), Cullen Fellow (2021), Physics TA Fellow (2024-25)
Advisor: Prof. Mark B. Wise · Thesis: *"Tracing New Physics: From Symmetry to Observation"*

M.Sc. & B.Sc. in Physics *2015–2020*
Sapienza University of Rome | 110/110 summa cum laude

Class Rank: #1 BSc (30.68/30), Top 5% MSc (30.32/30) · Sapienza School for Advanced Studies
European fellowships totaling €15,000+, including the Enrico Persico and INFN–NSF/LIGO fellowships

ACADEMIC APPOINTMENTS

Postdoctoral Scholar *Sept. 2026*
University of California, San Diego

Supervisor: Prof. Aneesh V. Manohar. Research in theoretical particle physics and cosmology.

Postdoctoral Scholar Research Associate *July–Aug. 2026*
California Institute of Technology

RESEARCH IMPACT

Theoretical Particle Physics and Cosmology *2020–2026*
Walter Burke Institute for Theoretical Physics (Caltech)

Studied warm inflation, leptogenesis in automatic Nelson–Barr models, galaxy-bias renormalization, axion-like particle production in beam-dump electromagnetic showers, and finite naturalness in quark–lepton unification. Combined analytical calculations and numerical simulations, yielding publications in PRD, JHEP, and JCAP.

Gravitational-Wave Phenomenology *2019–2020*
LIGO Lab (Caltech) & Virgo (Sapienza Università di Roma)

Developed a Bayesian framework to extract gravitational-wave polarization content and test General Relativity. The resulting Master's thesis received the 2020 Con.Scienze national prize.

INDUSTRY EXPERIENCE

Human Frontier Collective Specialist – Generative AI *May 2025-Present*
Scale AI | part-time

Designed and validated complex physics problems – from graduate level to frontier – for training LLMs. Ensured scientific rigor of AI responses in physics, bridging my expertise with practical AI applications.

PUBLICATIONS & TALKS

Selected Publications:

- S. Patrone et al., "Long-lived axionlike particles from electromagnetic cascades," *Phys. Rev. D* (2026)
- C. Murgui and S. Patrone, "Leptogenesis in automatic Nelson–Barr models," *JHEP* (2025)
- P. Fileviez Pérez et al., "Finite naturalness and quark–lepton unification," *Phys. Rev. D* (2024)
- S. Patrone et al., "Regularization-scheme dependence of the counterterms in the galaxy-bias expansion," *JCAP* (2023)

As a member of the LIGO Scientific Collaboration and Virgo Collaboration from 2019 to 2021, I co-authored 10 additional peer-reviewed collaboration papers. A complete publication list is available on iNSPIRE.

Invited Talks:
2026: UC Santa Cruz | 2025: Zurich, Texas A&M, Northwestern | 2024: CERN, Case Western

RECOGNITION

James A. Cullen Fellow 2021
Caltech academic excellence

Con.Scienze Prize 2020
National award for Master's thesis research

Danny Koh Fellow 2020
$18,000 graduate support

Enrico Persico Scholar 2020
€6,000 from Accademia dei Lincei

INFN/LIGO-NSF Fellow 2019
€5,000 for international research collaboration

TECHNICAL SKILLS

Theory: QFT, General Relativity, Cosmological Perturbations

Phenomenology: Cross Sections, Event Generation, Signal Analysis

Computing: Python, C, C++, Mathematica, ROOT

Methods: Adaptive Monte Carlo, Bayesian Inference, Boltzmann Equations, Loop Calculations

LANGUAGES

English (Fluent), Italian (Native), Spanish (Intermediate), Mathematics (Universal)

TEACHING & LEADERSHIP

Physics TA Fellow (2024-25)
Coordinated training and support for 100+ TAs; managed a 1.8K program budget and developed TA resources.

Teaching Excellence
100+ students, Certificate of Practice in University Teaching (Caltech)

Science Communicator (2018-20)
3 years bilingual museum tours (Sapienza)

Civic Leader (2015-18)
Youth Council President for 30K residents

BEYOND PHYSICS

Pianist · *Tiramisseur*